

WT32-SC01 Specification Sheet



Version

V3.2 Qiming

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Historical versions			
Version	Author	time	describe
1.0	YOUNG	2020-01-25 First creation; 2020-04-10 Modified	
3.2	YOUNG	physical product images and added board test parameters.	

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I. Overview

The WT32-SC01 is a development board designed for visual touchscreens. The board is equipped with a self-developed GUI platform firmware and supports...

The WT32-SC01 development board features a graphical drag-and-drop programming interface to help users develop custom control platforms. The main controller of the WT32-SC01 development board uses...

The ESP32-WROVER-B module is a general-purpose Wi-Fi+BT+BLE MCU module with 4MB of internal SPI Flash.

It also features 8MB of PSRAM. The WT32-SC01 development board can also be used for button operation, voice control, and camera recording via expansion interfaces on both sides.

The development and debugging of top-tier functions greatly shortens the user's development cycle.

II. Board Dimensions

The image below shows the dimensions of the WT32-SC01 development board. The board measures 58mm x 91mm and has four pins.

Each part has a positioning hole with a diameter of 4.05mm.

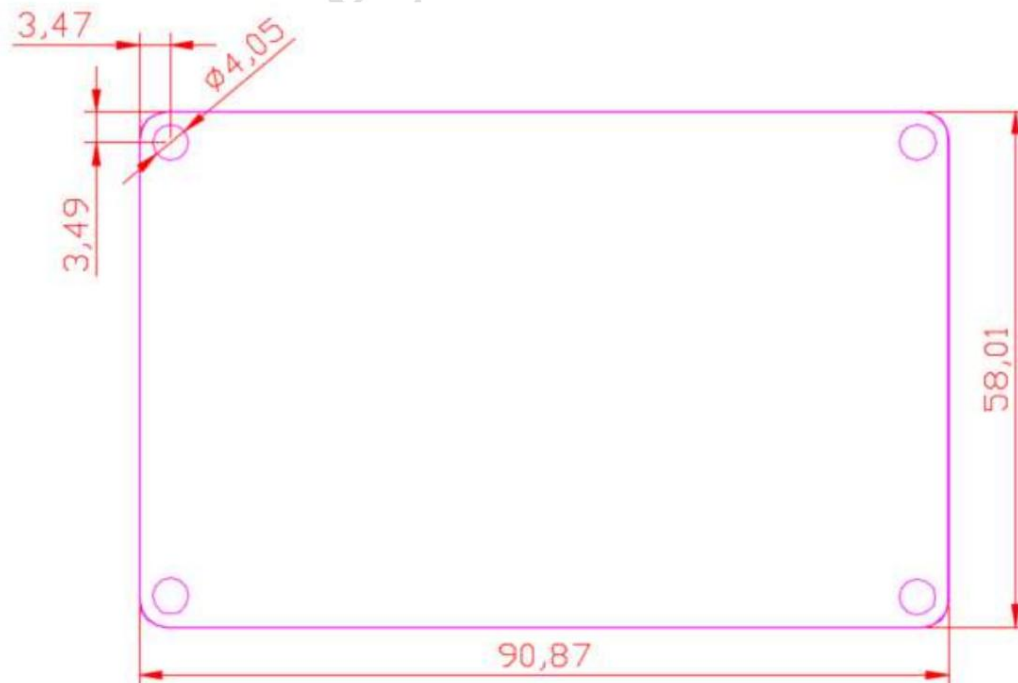


Figure 1-1: PCB Dimension Diagram



Figure 1-2: Schematic diagram of the actual dimensions of the circuit board (length and width)

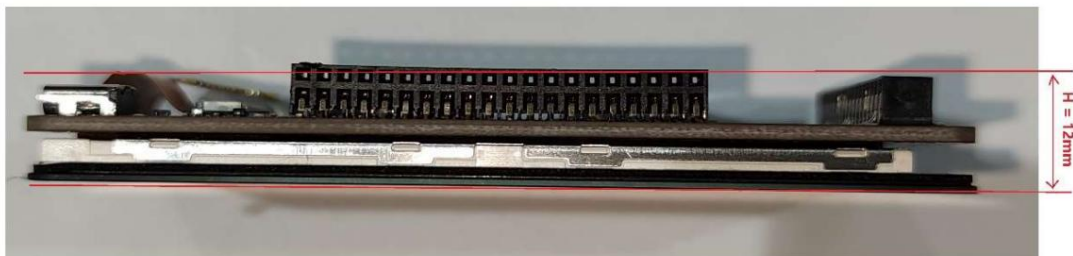


Figure 1-3: Schematic diagram of the actual dimensions of the circuit board (height)

III. Hardware Resources

The following is a physical image of the development board. The hardware resources of the development board are as follows:

Figure 2 shows the front view of the development board, which includes a 3.5-inch LCD screen with a resolution of 320x480 and a capacitive touch screen.

The template supports two-point

touch. Figure 3 shows the back of the development board, and its hardware resources are as follows:

1. ESP32-WROVER-B Module

The ESP32-WROVER-B is a powerful Wi-Fi+BT+BLE MCU module designed for various applications.

It has a wide range of applications, including low-power sensor networks and demanding applications such as voice coding, audio streaming, and MP3 decoding.

The specifications for this ESP32-WROVER-B are: 8Mbyte PSRAM + 4Mbyte Flash.

2. TP Interface

Capacitive touchpad interface, I2C interface, 0.5mm pitch, 6Pin, flip-top bottom connection.

3. LCM display interface

3.5-inch LCM display interface, SPI interface, 0.5mm pitch, 24 pins. This SPI interface connects to the ESP32.

It features a hardware HSPI interface with a clock frequency of up to 80MHz.

4. Power Management

Two 3.3V output LDOs, one powering the board itself and the other powering external expansion boards, with separate power supplies.

This is to prevent the expansion board from interfering with the power supply of the ESP32, thus ensuring the stable operation of the ESP32.

5. External power interface

The reserved external power supply interface (positive and negative as shown by the + and - signs in the diagram above) has an input voltage range of 5V-9V and requires a maximum load voltage of [missing information].

Flow I>2A

6. RST button

Connect the self-reset button to the EN pin of the ESP32. This button can be used to reset the ESP32.

7. Type-C interface

The universal USB-C interface (Type-C interface) is used for powering the development board, UART communication, and firmware downloading.

The download circuit hardware implements data flow control, so firmware download supports one-click automatic download.

8-9. Expansion Board Interface

The interface of the external expansion board allows for power supply, communication, control, and other functional expansion to meet user needs.

Multiple requirements. Interfaces 8 and 9 are pin-to-pin connections, so the expansion board plugs into interfaces 8 and 9...

The routing connections are the same; the only difference is that it provides a better user experience by allowing expansion boards to be installed in different orientations. Also: when the I/O of two expansion boards...

When there are no conflicts, two expansion boards can be plugged in simultaneously to achieve two expansion functions. Expansion board interface specifications: 2.0mm pitch.

2x20 pins, optional side-mounted female connectors. See Figure 3 - Extended Interface Definition Diagram for interface definition.

10. Power indicator light

The power indicator light (red light) will light up when the USB cable is plugged in.

11. UART communication indicator light

The TXD and RXD indicator lights in the UART will flash when there is data flow.

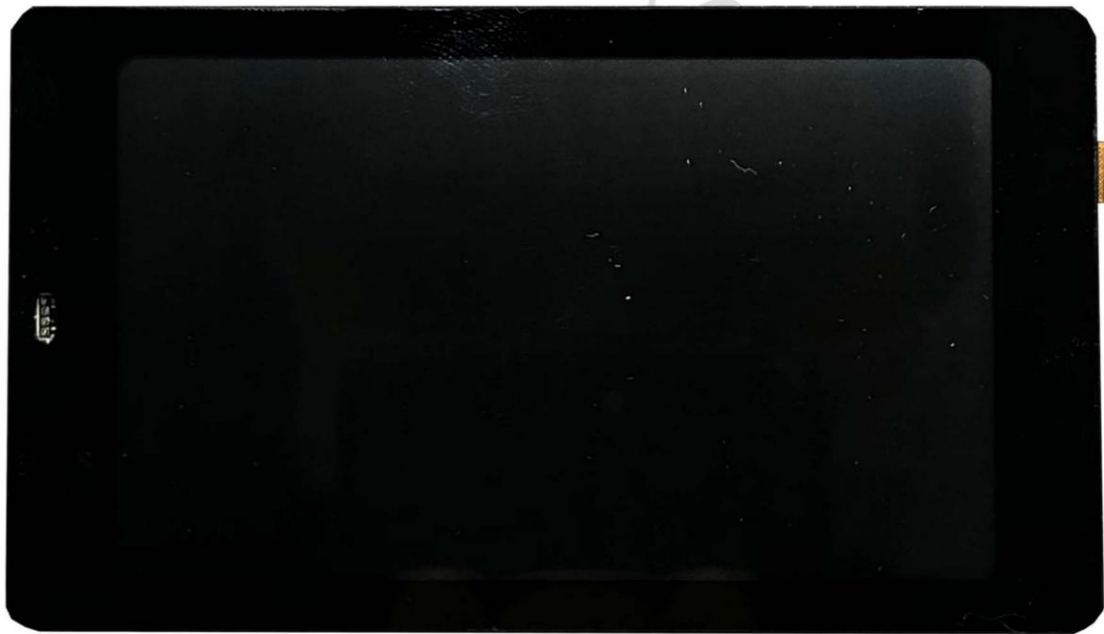


Figure 2: Front view of the WTPHIMainBoard development board



IV. Power-on Instructions

1. Power supply

This development board supports USB Type-C 5V power supply and has a reserved external power input interface (as shown in Figure 3, label 5).

When no other expansion cards are inserted, the recommended input voltage is 5V and the supported current is not less than 1A. When other expansion cards are inserted...

Recommended input voltage: 5V; Supported current: at least 2A (actual current should be determined based on the expansion board's power consumption). Note: (This is a direct translation of the original text, which is incomplete and requires further context.)

When the reserved power interface is used for external power supply, the power input voltage range is 5V-9V, and the recommended load current $I > 2A$.

2. Precautions

Please ensure that the FPC cables for the TP and LCM screens are plugged into the corresponding FPC sockets before powering on the development board.

This may damage the TP and LCM screens. If the

development board malfunctions, please first test the three TestPad points on the board (PP5V, PP3V3_1, and...).

Check if the voltage of PP3V3_2 is normal. PP3V3_1 outputs a 3.3V power supply to this board, while PP3V3_2 outputs a...

The external expansion board is powered by 3.3V, while PP5V is the 5V power supply for the USB input, which also powers the external expansion board. See Figure 5.

A schematic diagram of the power supply test points on the development board is shown. The voltage to ground at these three points is tested respectively.

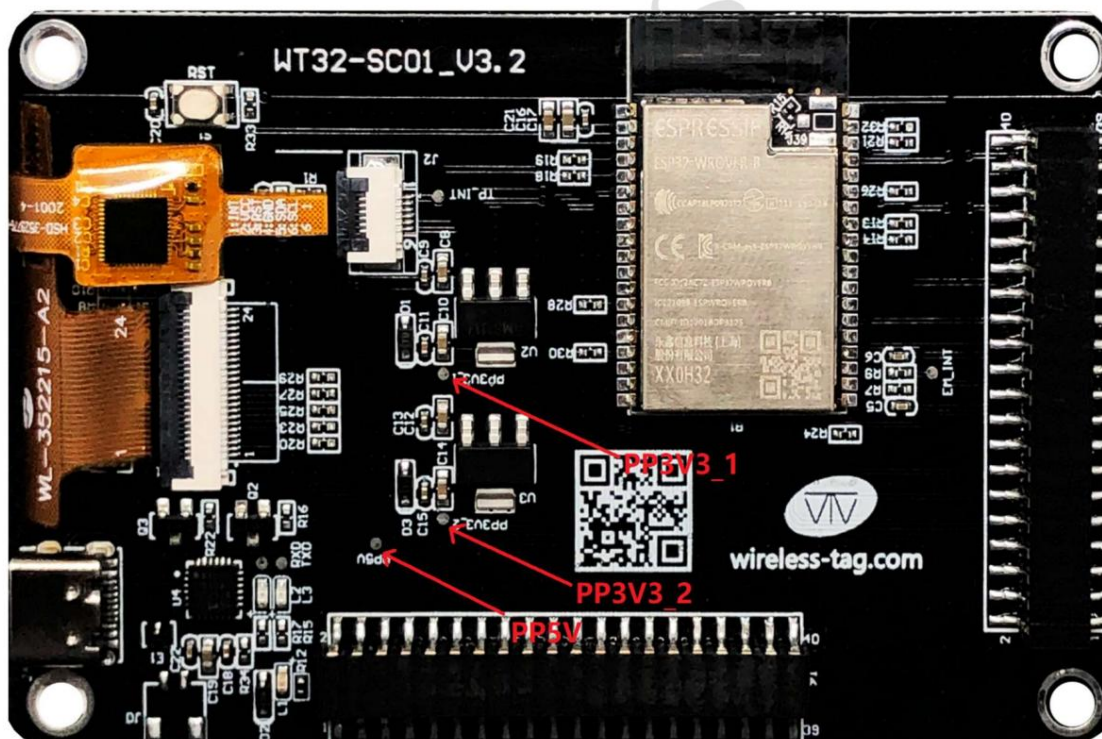


Figure 5: Schematic diagram of power test points on the development board

V. Circuit Board Electrical Parameters

Test

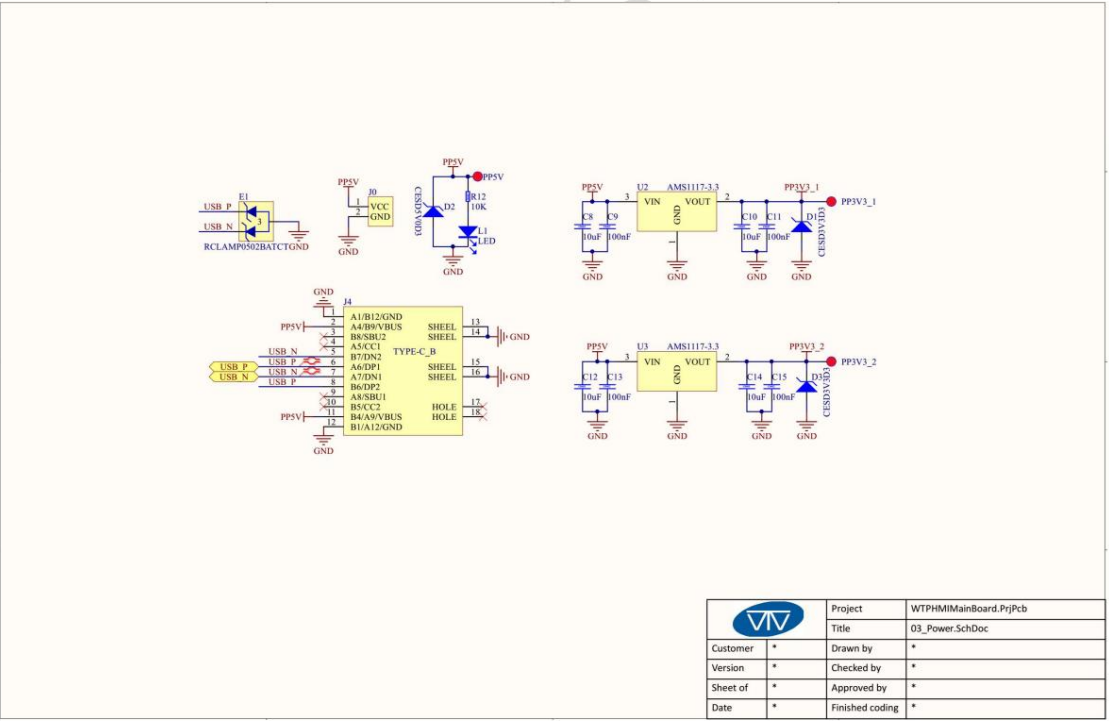
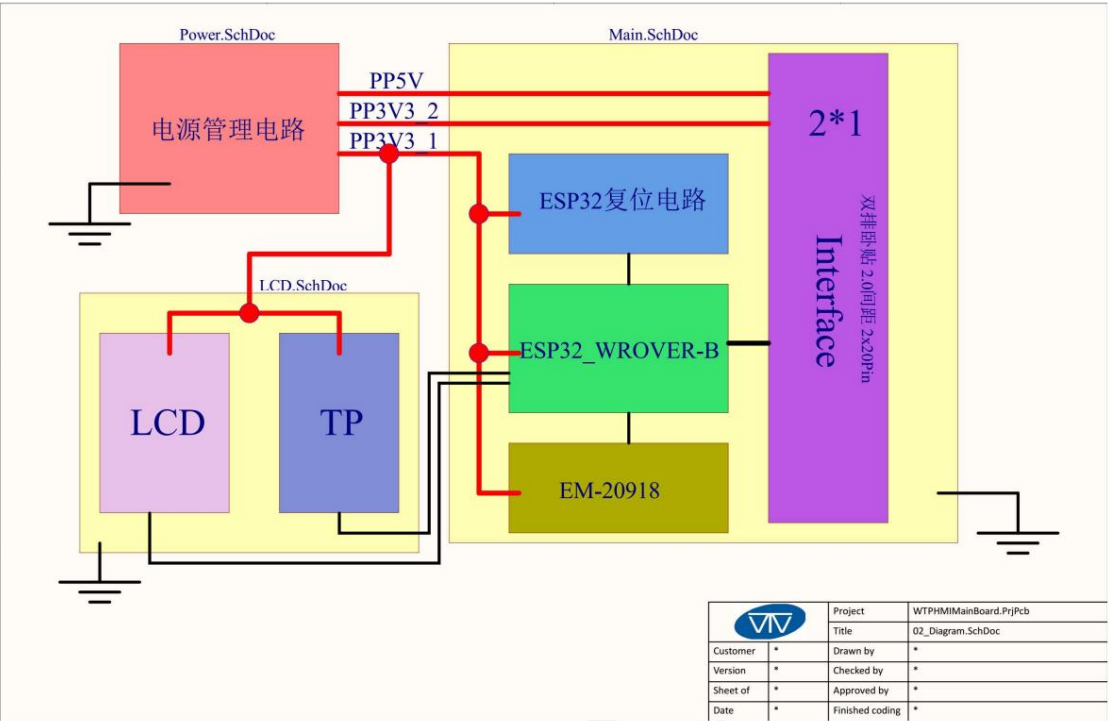
Environment: 1. Temperature:

Ambient temperature - 25° 2. Power Supply: 5.0V ripple V < 100mA

Test parameters:

1. Normal operating current: 240mA (average current)
2. Sleep mode operating current: less than 18mA (average current)

VI. Circuit Schematic





VII. Function Development

8. Firmware Burning

For details, please see <http://doc.8ms.xyz/docs/gui/gui-1bv0v24m8j8t7>

wireless-tag